評分準則 / Marking Scheme

2026 · Paper Paper 1 · 105 marks

通用評分規則

- M / A / PROOF 分類
- **f.t.**： M marks always follow through. A marks follow through ONLY when isFollowThrough = true.
- 扣分上限： A(1)=1 · A(2)=1 · B=0 (u, pp)
- Do not deduct in any step where the candidate scores 0.
- A correct answer merits ALL the marks of that part unless a particular method is specified in the question.
- Use of notation different from the scheme is NOT penalized. · Benefit of doubt
- All fractional answers must be simplified.

甲部(1) (35 marks)

1. 指數律 [B-]

(a) (3 分)

M1 $(x^{-2}y)^4 = x^{-8}y^4$

for expanding the power / 展開次方

M1 $\frac{x^5y^{-3}}{x^{-8}y^4} = x^{5-(-8)}y^{-3-4} = x^{13}y^{-7}$ f.t.

for subtracting indices / 相除時指數相減

A1 $= \frac{x^{13}}{y^7}$

positive indices required / 必須以正指數表示

2. 公式變換 [B-]

(a) (3 分)

M1 $3h + 5 = k(h-2)$

for cross multiplication / 交叉相乘

M1 $3h - kh = -2k - 5 \Rightarrow h(3-k) = -(2k+5)$ f.t.

for grouping the h terms / 集合含 h 的項

A1 $h = \frac{2k+5}{k-3}$

or equivalent

3. 估算與誤差 [B-]

(a) (1 分)

A1 $250 - \frac{10}{2} = 245$ g

u-1 if unit g is missing / 漏寫單位 g 扣 1 分

(b) (2 分)

M1 最大可能總重量 $= 4 \times 255 = 1020$ g f.t.

for using the upper limit 255 g / 使用上限 255 g

A1 $1024 > 1020 \Rightarrow$ 故不可能 f.t.

1A f.t. — conclusion must be stated / 結論句必須寫出

4. 因式分解 [B]

(a) (2 分)

M1 $25m^2 - 20mn + 4n^2 = (5m)^2 - 2(5m)(2n) + (2n)^2$

for recognizing the perfect square / 辨認完全平方式

A1 $= (5m - 2n)^2$

(b) (2 分)

M1 $= (5m - 2n)^2 - 3(5m - 2n)$ f.t.

for using the result of (a) / 引用 (a) 的結果

A1 $= (5m - 2n)(5m - 2n - 3)$

pp-1 if not fully factorized / 未完全分解扣 1 分

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5. 百分法 [B]

(a) (4 分)

M1 設成本為\C。$25\% \times C = 260$

for setting up the profit equation / 建立利潤方程

A1 $C = 1040$

M1 售價 $= 1040 + 260 = 1300$；標價 $\times (1 - 20\%) = 1300$ f.t.

for using selling price = MP x 80% / 售價 = 標價 x 80%

A1 標價 $=$ \1625

u-1 if the monetary unit is missing / 漏寫 $ 扣 1 分

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6. 一元一次不等式 [B]

(a) (3 分)

M1 $-3x - 6 > x - 14 \Rightarrow -4x > -8 \Rightarrow x < 2$

sign must be reversed on dividing by -4 / 除以負數須反轉不等號

M1 $2x + 5 \leq 1 \Rightarrow x \leq -2$

A1 取並集：$x < 2$ f.t.

for 'or' the union is taken / 「或」須取並集

(b) (1 分)

A1 1 f.t.

1A f.t. from (a)

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7. 坐標變換與斜率 [B+]

(a) (2 分)

A1 $A' = (8, 6)$

clockwise 90 degrees: (x, y) -> (y, -x)

A1 $B' = (-5, -2)$

(b) (2 分)

M1 斜率 $= \frac{-2-6}{-5-8}$ f.t.

1M f.t. from (a)

A1 $= \frac{8}{13}$

fraction must be simplified / 分數須化至最簡

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8. 全等三角形 [B+]

(a) (2 分)

PROOF $AB = AD$（已知）；$\angle BAC = \angle DAC$（已知）；$AC = AC$（公共邊）
Case 1: correct proof WITH reasons = 2; Case 2: correct proof WITHOUT reasons = 1

PROOF $\therefore \triangle ABC \cong \triangle ADC$ （邊、角、邊）
the reason SAS must be quoted for the 2nd mark / 須寫出理由「邊、角、邊」

(b) (3 分)

M1 $\angle ADC = \angle ABC = 116^\circ$ （全等三角形的對應角） f.t.
for using the result of (a) / 引用 (a) 的結果

M1 $\angle BAD + \angle ABC + \angle BCD + \angle ADC = 360^\circ$
angle sum of a quadrilateral / 四邊形內角和

A1 $\angle BAD = 360^\circ - 116^\circ - 116^\circ - 84^\circ = 44^\circ$
u-1 if the degree symbol is missing / 漏寫「度」扣 1 分

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9. 統計圖表與概率 [B+]

(a) (1 分)

A1 $21 - 25$分鐘

(b) (2 分)

M1 平均數 $= \frac{4(13)+10(18)+14(23)+8(28)+4(33)}{40} = \frac{910}{40}$
for using class marks / 使用組中點

A1 $= 22.75$分鐘
u-1 if the unit is missing / 漏寫單位扣 1 分

(c) (2 分)

M1 $P = \frac{4+10}{40}$

A1 $= \frac{7}{20}$
fraction must be simplified / 分數須化至最簡

甲部(2) (35 marks)

10. 變分 [A-]

(a) (3 分)

M1 $f(x) = a + bx^2$，其中$b \neq 0$
for the correct form of partial variation / 部分變分的正確形式

M1 $\begin{cases} a + b = 13 \\ a + 16b = 58 \end{cases} \Rightarrow 15b = 45$ f.t.
for forming the simultaneous equations / 建立聯立方程

A1 $b = 3$，$a = 10 \Rightarrow f(x) = 10 + 3x^2$
for both a and b correct / 兩者皆正確方可得分

(b) (1 分)

A1 10 f.t.
1A f.t. from (a)

(c) (3 分)

M1 $10 + 3x^2 = k \Rightarrow x^2 = \frac{k-10}{3}$ f.t.

M1 有兩個不同的實根 $\Leftrightarrow \frac{k-10}{3} > 0$ f.t. alt: q10c_alt
alternatively use discriminant of 3x^2 + (10-k) = 0 : 0 - 4(3)(10-k) > 0

A1 $k > 10$

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11. 統計數據的量度 [A-]

(a) (3 分)

M1 中位數 $= \frac{(20+a)+28}{2} = 27$

median = mean of the 6th and 7th data / 中位數為第 6 及第 7 個數據的平均數

M1 $Q_1 = \frac{19+22}{2} = 20.5$，$Q_3 = \frac{(30+b)+36}{2}$，$Q_3 - Q_1 = 15$ f.t.

for setting up the IQR equation / 建立四分位數間距方程

A1 $a = 6$，$b = 5$

for both correct / 兩者皆正確方可得分

(b) (2 分)

A1 平均數 $= \frac{336}{12} = 28$ f.t.

A1 標準差 $= \sqrt{88} \approx 9.38$ f.t.

1A r.t. 9.38

r.t. 9.38

(c) (2 分)

M1 新中位數 $=$ 餘下11個數據的第6個 $= 26$ f.t.

A1 中位數減少$27 - 26 = 1$ f.t.

1A f.t. — the direction of change must be stated / 必須說明增減方向

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12. 多項式的餘式定理與因式定理 [A]

(a) (3 分)

M1 $p(x) = (x^2 - x + 4)(2x + t) + x - 13$

for the division algorithm with linear quotient 2x + t / 除法算式，商為 2x + t

M1 常數項：$4t - 13 = -33 \Rightarrow t = -5$ f.t.

for comparing coefficients / 比較係數

A1 $a = t - 2 = -7$，$b = 8 - t + 1 = 14$

for both correct / 兩者皆正確方可得分

(b) (2 分)

M1 $p(3) = 2(27) - 7(9) + 14(3) - 33$ f.t.

for using the factor theorem / 運用因式定理

A1 $p(3) = 54 - 63 + 42 - 33 = 0 \Rightarrow x - 3$是$p(x)$的因式 f.t.

1A f.t. — the conclusion must be written / 必須寫出結論句

(c) (2 分)

M1 $p(x) = (x - 3)(2x^2 - x + 11)$；$\Delta = (-1)^2 - 4(2)(11) = -87$ f.t.

1M for the factorization OR for the discriminant / 因式分解或求判別式皆得分

A1 $\Delta < 0 \Rightarrow 2x^2 - x + 11 = 0$沒有實根。$p(x) = 0$只有一個實根$x = 3$，且為有理數 $\Rightarrow$ 不同意 f.t.

1A f.t. — conclusion sentence must be written / 結論句必須寫出

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13. 相似立體圖形的體積與表面積 [A]

(a) (3 分)

M1 較大圓錐的體積 $= \frac{1}{3}\pi(12)^2(18) = 864\pi \text{ cm}^3$
for the volume formula of a cone / 圓錐體積公式

M1 $\frac{V_{較小}}{864\pi} = \frac{8}{27}$ f.t.
for using the ratio of volumes / 運用體積比

A1 $V_{較小} = 256\pi \text{ cm}^3$
u-1 if the unit cm^3 is missing / 漏寫單位 cm^3 扣 1 分

(b) (2 分)

M1 $\pi(8)^2h = 864\pi + 256\pi$ f.t.
for conservation of volume / 體積守恆

A1 $h = 17.5 \text{ cm}$
u-1 if the unit cm is missing / 漏寫單位 cm 扣 1 分

(c) (2 分)

M1 圓柱總表面積 $= 2\pi(8)^2 + 2\pi(8)(17.5) = 408\pi \approx 1281.8 \text{ cm}^2$ f.t.
1M f.t. from (b)
r.t. 1282

A1 兩圓錐總表面積之和 $\approx 583\pi \approx 1831 \text{ cm}^2$；$408\pi < 583\pi \Rightarrow$ 同意 f.t.
1A f.t. — slant heights: 6\sqrt{13} and 4\sqrt{13}
r.t. 1831

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14. 圓的方程 [A]

(a) (2 分)

A1 $G = \left(-\frac{-24}{2}, -\frac{-10}{2}\right) = (12, 5)$

A1 $r = \sqrt{12^2 + 5^2 - 120} = \sqrt{49} = 7$

(b) (1 分)

A1 $GH = \sqrt{(18-12)^2 + (13-5)^2} = \sqrt{100} = 10$ f.t.

(c)(i) (1 分)

A1 $GP \perp GH$
accept 'GP is perpendicular to GH' / 亦接受「GP 垂直於 GH」

(c)(ii) (3 分)

M1 $GP = r = 7$，$GH = 10$，$\angle HGP = 90°$ f.t.

M1 $HP = \sqrt{7^2 + 10^2} = \sqrt{149}$ f.t.
for Pythagoras' theorem / 畢氏定理

A1 周界 $= 7 + 10 + \sqrt{149} \approx 29.2$
1A r.t. 29.2
r.t. 29.2

乙部 (35 marks)

15. 排列、組合與概率 [A]

(a) (3 分)

M1 總結果數目 $= C_4^{15} = 1365$

M1 有利結果 $= C_2^6 \times C_2^9 = 15 \times 36 = 540$ f.t.

A1 $P = \frac{540}{1365} = \frac{36}{91}$

fraction must be simplified / 分數須化至最簡

(b) (3 分)

M1 $P = 1 - P(\text{沒有紅球}) - P(\text{沒有黃球}) + P(\text{既沒有紅球亦沒有黃球})$

for the inclusion-exclusion principle / 容斥原理

M1 $= \frac{1365 - C_4^9 - C_4^{11} + C_4^5}{1365} = \frac{1365-126-330+5}{1365}$ f.t.

A1 $P = \frac{914}{1365}$

1A r.t. 0.670

r.t. 0.670

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16. 二次函數的圖像 [A]

(a) (2 分)

M1 $f(x) = 2(x^2 - 4kx) + 10k^2 + 12k + 6 = 2(x-2k)^2 - 8k^2 + 10k^2 + 12k + 6$

for completing the square / 配方法

A1 頂點 $= (2k\text{，}2k^2 + 12k + 6)$

(b) (2 分)

M1 $2k^2 + 12k + 6 = 2(k+3)^2 - 12$ f.t.

for completing the square again / 再次配方

A1 最小值 $= -12$，此時$k = -3$

for both correct / 兩者皆正確方可得分

(c) (2 分)

M1 把$x = 2k$代入Γ：$y = \frac{1}{2}(2k)^2 + 6(2k) + 6 = 2k^2 + 12k + 6$ f.t.

1M f.t. from (a)

A1 此值等於頂點的y坐標 $\Rightarrow$ 同意 f.t.

1A f.t. — conclusion must be stated / 結論句必須寫出

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17. 等差數列與等比數列 [A]

(a) (2 分)

M1 $d = \frac{53-23}{9-3} = 5$

for finding the common difference / 求公差

A1 $A(n) = 23 + (n-3)(5) = 5n + 8$

(b) (1 分)

A1 $S(50) = \frac{50}{2}[13 + 258] = 6775$ f.t.

(c) (3 分)

M1 $G(1) = A(2) = 18 \Rightarrow \sum_{i=1}^{n} G(i) = \frac{18(2^n - 1)}{2-1} = 18(2^n - 1)$ f.t.

for the sum of a geometric sequence / 等比數列求和公式

M1 $18(2^n - 1) > 6775 \Rightarrow 2^n > \frac{6775}{18} + 1 \approx 377.4 \Rightarrow n > \log_2 377.4 \approx 8.56$ f.t.

accept solving by trial with n = 8 and n = 9 / 亦接受試值法

A1 最小的$n = 9$　（$n = 9$：$9198 > 6775$；$n = 8$：$4590 < 6775$）

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18. 立體圖形的三角學 [A+]

(a) (2 分)

M1 $OM = \frac{16}{2} = 8$；$VM^2 = VO^2 + OM^2 = 15^2 + 8^2$
for Pythagoras' theorem in right-angled triangle VOM / 在直角三角形 VOM 中用畢氏定理

A1 $VM = \sqrt{289} = 17$ cm

(b) (2 分)

M1 所求的角 $= \angle VMO$；$\tan\angle VMO = \frac{VO}{OM} = \frac{15}{8}$ f.t.
1M for identifying the correct angle / 正確辨認二面角

A1 $\angle VMO \approx 61.9°$
1A r.t. 61.9 degrees
r.t. 61.9

(c) (4 分)

M1 $VM \perp BC$及$VN \perp AD$；$BC\ //\ AD \Rightarrow$ 所求的角 $= \angle MVN$
1M for identifying the angle between the two planes / 辨認兩平面的夾角

M1 $VM = VN = 17$ cm，$MN = 16$ cm f.t.

M1 $\cos\angle MVN = \frac{17^2+17^2-16^2}{2(17)(17)} = \frac{322}{578}$ f.t.
for the cosine formula / 餘弦公式

A1 $\angle MVN \approx 56.1° < 60° \Rightarrow$ 同意 f.t.
1A f.t. r.t. 56.1 degrees — conclusion must be stated
r.t. 56.1

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19. 圓的方程與切線 [A+]

(a) (2 分)

A1 $G = (6, 8)$

A1 $r = \sqrt{6^2 + 8^2 - 0} = 10$

(b) (3 分)

M1 G至L的距離 $= \frac{|3(6)+4(8)-100|}{\sqrt{3^2+4^2}} = \frac{50}{5} = 10$
for the point-to-line distance formula / 點到直線距離公式

PROOF 距離 $= 10 = r \Rightarrow L$是C的切線 f.t.
conclusion required / 必須寫出結論

A1 $T = (12, 16)$
T is the foot of perpendicular from G to L / T 為 G 至 L 的垂足

(c) (2 分)

M1 $0^2 + 0^2 - 12(0) - 16(0) = 0 \Rightarrow O$在$C$上；$OT$的中點 $= \left(\frac{0+12}{2}, \frac{0+16}{2}\right) = (6, 8) = G$ f.t.
alt: q19c_alt
alternatively show OT = 20 = 2r / 亦可證 OT = 20 = 2r

A1 OT的中點為圓心$G \Rightarrow$ 同意，OT是直徑 f.t.
1A f.t. — conclusion must be stated

(d) (2 分)

M1 以OT為底，最大的高 $= r = 10$ f.t.
the height is greatest when P is an end-point of the diameter perpendicular to OT

A1 $S_{\max} = \frac{1}{2}(20)(10) = 100$
